
5354SMFT6Y: STATISTICAL METHODS FOR THE PHYSICAL SCIENCES

UNIVERSITY OF AMSTERDAM

SYLLABUS, JANUARY 2026

LOGISTICS:

Lectures/Class:

- Monday, 9:00 - 13:00 (weeks 2-5)
- Wednesday, 9:00 - 13:00 (weeks 2-4)
- Friday, 13:00 - 17:00 (weeks 2-4)
- Tuesday, 13:00 - 17:00 (week 5)
- **Exam: Friday, Jan 30, 9:00 - 12:00 (Exam will be 2 hours long, 2h30 for students with extra time)**

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COURSE DESCRIPTION

Data is everywhere! It now plays a vital part in nearly every aspect of human decision making and is just as important in scientific research. This is especially true in physics and astronomy, where our understanding depends heavily on observations of the natural world. But how do we turn raw data into meaningful knowledge about nature? How can we make sense of the claims we encounter in scientific literature, or in the news? This is where statistics comes in. Statistics, grounded in the mathematics of probability and logic, provides the tools we need to extract insights from data. These methods are not only central to academic research but are also fundamental in a wide range of careers, from data science to finance.

This course offers an intensive introduction to statistical methods and probabilistic thinking, with a focus on interpreting data in astronomy and the physical sciences. While rooted in scientific research, the skills gained have wide-reaching applications beyond it. The course begins by establishing a solid mathematical foundation, providing the tools to adapt statistical methods to new situations. Building on this base, it emphasises intuition and hands-on experience over abstract proofs, focusing on practical skills that can be applied right away. Through intensive training with Python-based data analysis challenges, you'll learn to work with real data from physics and astronomy and gain a solid understanding of where and how to apply statistical methods.

INTENDED LEARNING OUTCOMES

By the end of this course, students should be able to ...

- Explain the foundational concepts of statistics, including probability, randomness, statistical distributions.
- Formulate a statistically robust data analysis process that considers the limitations of the statistical methods used and the data to be analysed.
- Choose an appropriate statistical test for a given problem, and employ hypothesis testing to make inferences about data.
- Apply Bayesian probability theory to scientific problems.

- Develop simple computer programmes to analyse data and apply statistical testing in practice.
- Critically interpret the results of a statistical test, including an evaluation of the appropriateness of the test, and its assumptions and limitations, in comparison to the data.
- Evaluate scientific results in the academic literature within physics/astronomy and in popular media.

IMPORTANT DATES

Jan 5	First lecture
Jan 14	First homework due
Jan 27	Second homework due
Jan 30	Exam

GRADING

- 25% Homework #1
- 25% Homework #2
- 50% Final exam

TYPICAL (APPROXIMATE) CLASS FORMAT

- 2 x 45-minute interactive lectures
 - Interactive lectures introducing new material and providing immediate practice opportunities
- 2 x 45-minute tutorial sessions
 - 30 minutes: practice exam questions
 - 15 minutes: information/background about assignments
 - 45 minutes: interactive help with assignments

(APPROXIMATE) SCHEDULE

Class Dates	Due	Mon	Tue	Wed	Fri
Jan 5, 7 and 9		Introduction + Statistical fundamentals (Episode 1 and 8)	-	Statistical fundamentals and probability theory (Episodes 1,2 and 8)	Statistical Distributions (Episode 2)
Jan 12, 14 and 16	Jan 14: Homework #1	Hypothesis testing (Episodes 3, 4 and 7)	-	Statistical inference with models (Episode 5)	Bayesian inference (Episode 9)
Jan 19, 21, 23		Sampling methods (Episodes 10 and 11)	-	Model criticism and model comparison	Statistics & AI
Jan 26, 27, 30	Jan 27: Homework #2	Lying with statistics (and why you shouldn't)	Revision		Exam

LATE ASSIGNMENTS

Assignments are due Jan 14 (12:00pm / noon) for homework #1 and Jan 27 (12:00 pm / noon) for homework #2.

Late work which is submitted up to 24 hours after the deadline will receive a penalty of -20% of the awarded grade, while work submitted 24-48 hours after the deadline will receive a penalty of -40% of the awarded grade. Work submitted later than this will not be assessed. The rubric for the assessment of all the assignments, listing the categories assessed and the requirements for each of them, will be provided separately on Canvas.

RECOMMENDED READING

There is no required textbook for this course. There are online lecture notes that accompany this course (<https://huppenkothen.org/stats-methods-26/>).

Note: the online lecture notes provide significant background information and support the assignments, but don't map onto the in-class lectures 100%. Attending the lectures is strongly recommended!

A significant part of the course loosely follows the textbook:

Simon Vaughan, *Scientific Inference: Learning from Data*, Cambridge University Press, 2013, ISBN 9781107607590.

Additional resources that may be helpful:

- Ivezic, Connolly, VanderPlas, & Gray, *Statistics, Data Mining, and Machine Learning in Astronomy*, 2014, Princeton University Press
- Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin. *Bayesian Data Analysis*, 3rd Edition, 2013, Chapman & Hall/CRC Texts in Statistical Science.
- Hilbe, de Souza, and Ishida, *Bayesian Models for Astrophysical Data*, 2017, Cambridge University Press.

(GENERATIVE) AI POLICY

Over the past ~2 years, generative AI and especially Large Language Models like ChatGPT, Gemini and Claude have become common assistive tools in both research and education. In this course, you are encouraged¹ to experiment with generative AI to **aid your learning and understanding**, and to use it in ways that **support your work, not replace it**.

Learning the fundamentals of statistics is important even in the age of AI because for anything but the most trivial of applications, *AI makes catastrophic mistakes in ways that require expertise to spot* (believe me, I've tried!).

With this in mind, you are allowed and encouraged to use AI for this work in the following ways:

- To help you **implement code**: You may use AI to figure out the precise syntax for rote programming tasks, such as plotting and database manipulation, but **do not ask for and copy full solutions to assignment questions**
- to help you **debug code**: you may ask the AI for hints, explanations, or checks, to help you think through error messages, but **do not ask for or simply copy full solutions**

¹ But certainly not required! There are a wide range of very legitimate ethical and moral objections to the use of generative AI

- to help you understand concepts (e.g. for astrophysical quantities and processes you may not know) and think through mistakes, but AI cannot and must not decide for you which assumptions are reasonable, which tests are appropriate, or how to interpret results.
- **To generate practice questions for the exam** based on your notes, the lecture notes and slides, and to help you think through the solutions, spot mistakes and deepen your understanding.

You may **not** ask AI to generate full functions for you, or to provide full answers to the questions. **To do so is considered academic fraud and will be addressed as such.**

I encourage you to explore UvA's own UvA AI Chat (<https://aichat.uva.nl/>), which has important privacy protections for you that not available in commercial chatbots (for more information, see: <https://www.uva.nl/en/about-the-uva/about-the-university/ai/ai-in-education/uva-ai-chat/uva-ai-chat.html>). You are especially encouraged to experiment with the ChatBot's "StudyMode", which can help you clarify concepts and test your knowledge. If you use AI to help you study, make sure to cross-check the answers with a trusted source! AI can and will hallucinate information that looks right, but is not!

Note that you are responsible for the correctness of all work you submit, no matter its source. You must be able to explain your solutions and answers in detail if asked. You must flag any support by AI clearly and specifically (e.g. if it helped you figure out code, make sure you specify how in a docstring or text cell).

For more information, you can find UvA's policy framework on AI here: <https://www.uva.nl/en/about-the-uva/policy-and-regulations/education/policy-framework-and-guidelines-on-genai-in-education.html>